Pratham Modi

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Full Stack Engineer | Python, FastAPI, React, Next.js, PostgreSQL, Redis, AWS

Summary

Full-stack software engineer with 2+ years of production experience building high-performance web applications and distributed backends. Built and scaled multi-tenant SaaS platforms onboarding 10K+ users at C3iHub (IIT Kanpur) and AI-driven Next.js applications at Playpower Labs, spanning Python (FastAPI), Node.js, React/Next.js frontends, and event-driven data streaming (Kafka, Redis).

Technical Skills

Languages & Frameworks: Python, FastAPI, React, Next.js, Node.js, Express.js, TypeScript, JavaScript, SQL
Backend & Architecture: Microservices, RESTful APIs, WebSocket, Event-Driven Architecture, Distributed Systems, System Design
Databases & Caching: PostgreSQL, MongoDB, Redis, Redis Streams, BullMQ
Cloud & DevOps: AWS (EC2, S3, EKS), Docker, Kubernetes, CI/CD, GitHub Actions, OpenTelemetry, Prometheus, Grafana

Experience

C3iHub, IIT Kanpur Jul 2025 – Present
Software Development Engineer (Backend & Infrastructure) Bengaluru / Kanpur, India

- Led engineering of official IITK platform onboarding 10K+ participants pan-India; productized into multi-tenant SaaS serving public and private enterprise deployments with RBAC.
- Architected core platform backend and microservices using Express.js, MongoDB, Redis, and Socket.IO, serving 50K+ users and maintaining 99.9% uptime under 10,000+ concurrent connections.
- Engineered event-driven notification pipeline handling 75K+ daily events at 99.5% delivery reliability using Kafka, BullMQ, and Redis Pub/Sub with consumer-side deduplication.
- Built real-time collaborative round engine handling 2,000+ concurrent WebSocket sessions via Django Channels and Redis channel groups with race-condition-safe state synchronization.

Playpower Labs Nov 2024 – Jun 2025
Software Development Engineer (Full-Stack & AI Services) Bengaluru, India

- Developed AI tutoring platform using Next.js with SSR/ISR and React Server Components alongside FastAPI microservices, supporting 500+ concurrent WebSocket sessions with sub-100ms response times.
- Reduced release cycle time by 65% by implementing blue-green CI/CD on AWS with GitHub Actions and automated rollback triggers for zero-downtime deployments.
- Increased user session depth by 35% by integrating ChromaDB vector search over 10K+ documents via a RAG pipeline, achieving sub-200ms retrieval latency at p95.

Projects

Autonomous Invoice Processing Platform | *Kafka, Redis Streams, FastAPI, PostgreSQL, AWS*

- Engineered event-driven platform automating invoice intake across multi-channel APIs with automated validation, deduplication, and full audit trail.
- Built fault-tolerant microservices with consumer groups, dead-letter queues, exponential backoff retries, and RBAC-based human review workflow.

DeployMind: AI-Powered GitOps Platform | *Python, FastAPI, CrewAI, Redis, Docker, Kubernetes*

- Built GitOps platform automating deployments from GitHub to EC2 and Kubernetes (EKS) via a 3-agent AI pipeline using CrewAI with security gates.
- Implemented rolling, canary, and blue-green deployment strategies with Redis-backed queue, distributed locking, and 200+ automated tests.

Education

Pandit Deendayal Energy University 2021 – 2025
Bachelor of Technology in Computer Science and Engineering CGPA: 9.20 / 10.0